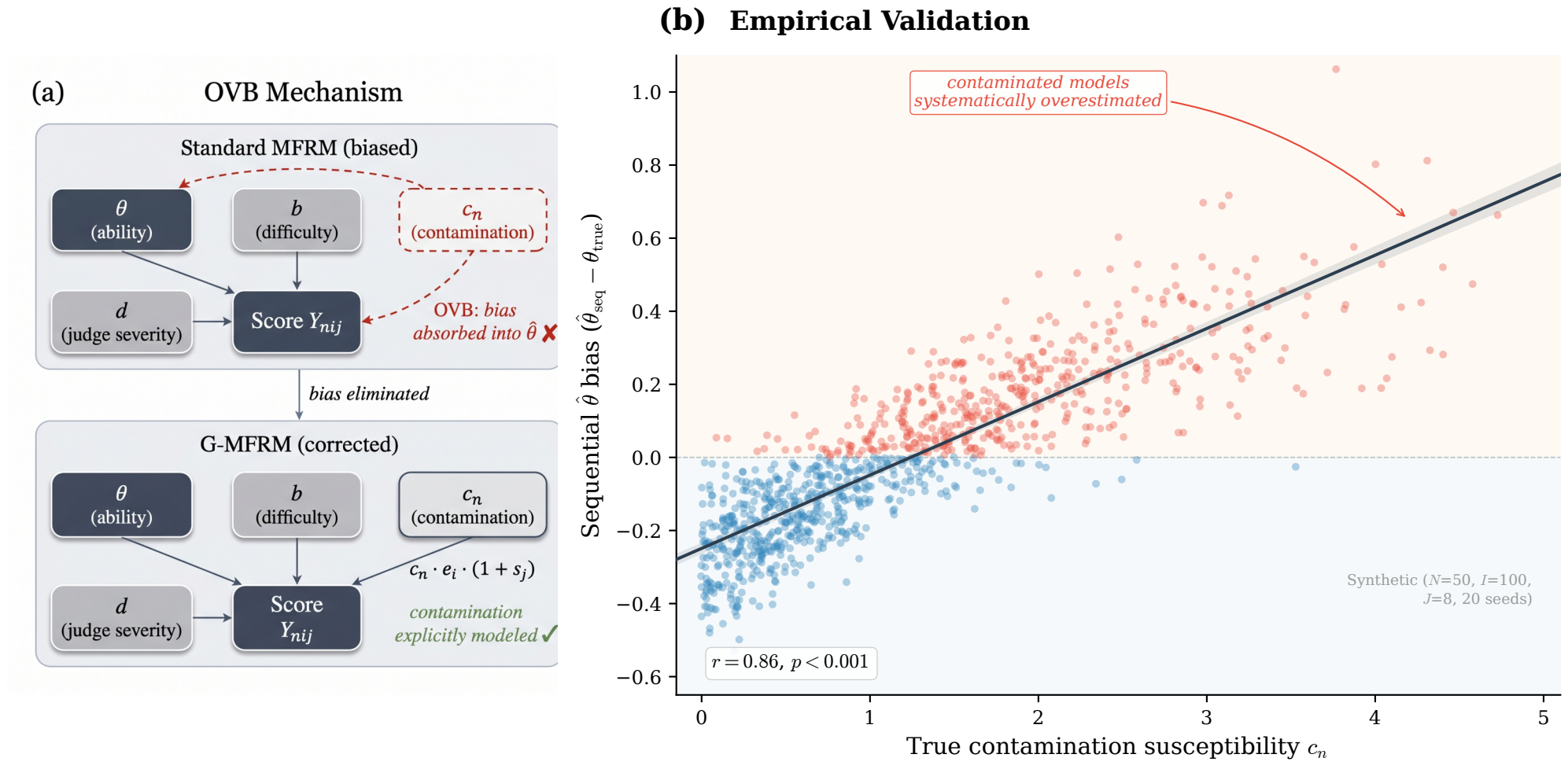

(a) OVB Mechanism
Standard MFRM (biased)
θ (ability)
b (difficulty)
c_n (contamination)
d (judge severity)
Score Y_{nij}
OVB: bias absorbed into $\hat{\theta}$ ✗
bias eliminated
G-MFRM (corrected)
θ (ability)
b (difficulty)
c_n (contamination)
$c_n \cdot e_i \cdot (1 + s_j)$
d (judge severity)
Score Y_{nij}
contamination explicitly modeled ✓
(b) Empirical Validation
contaminated models systematically overestimated
Sequential $\hat{\theta}$ bias ($\hat{\theta}_{\text{seq}} - \theta_{\text{true}}$)
True contamination susceptibility c_n
Synthetic (N=50, I=100, J=8, 20 seeds)
$r = 0.86$, $p < 0.001$